ITA0502-COMPUTER VISION

Activity-1

1. Medical Image Enhancement

A doctor receives an X-ray image in which the bones are visible, but the darker tissues have very poor visibility.

Questions:

1. Which gray-level transformation would you recommend?
2. Why is it suitable?
3. What effect will it have on the image?

2. Night-Time Surveillance

A CCTV camera captures a parking lot at night. Most pixel values are concentrated between **0 and 40**, making the image very dark.

Questions:

1. Which transformation would improve the visibility?
2. Why is it more suitable than negative transformation?
3. Predict the appearance of the enhanced image.

3. Fingerprint Recognition

A fingerprint image contains many fine ridge details, but the contrast between the ridges and valleys is very low.

Questions:

1. Which gray-level transformation would you apply?
2. Explain why.
3. What would happen if a log transformation were applied instead?

Activity - 1

1. Medical Image Enhancement:

Questions :-

1) which gray - - - - -

Ans) Power-law (Gamma) Transformation

2) why is it suitable?

Ans) It enhances dark regions without excessively affecting bright regions, making soft tissues more visible in the X-ray.

3) what effect - - - -

Ans) Dark tissue become brighter & clearer, while bones remain visible, improving the overall image quality.

2. Night - Time surveillance:

Questions

1) which - - - - - -

Ans) Log Transformation.

2) why is it - - - - - - - -

Ans) Log transformation expands low-intensity (dark) pixel values, making dark areas brighter. Negative transformation only inverts the image & does not improve visibility.

3) Predict the - - - - - -

Ans) The Parking lot will appear brighter, and objects such as vehicles, People, and surroundings will be easier to identify.

3. Fingerprint Recognition:

1) which gray - - - - - -

Contrast stretching (Piecewise-Linear Transformation)

2) Explain why.

It increases the contrast between fingerprint ridges and valleys, Making the fine ridge details more distinct for accurate recognition.

3) what happend - - - - - - -

Ans) Log transformation would brighten the darker regions but would not effectively enhance the ridge-to-valley contrast. As a result, fingerprint details may become less distinct compared to piecewise linear transformation.